FAST: Fast Audio Spectrogram Transformer

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***Abstract*—In audio classification, developing efficient and robust models is critical for real-time applications. Inspired by the design principles of MobileViT, we present FAST (Fast Audio Spectrogram Transformer), a new architecture that combines convolutional neural networks (CNNs) and transformers to capitalize on the strengths of both. FAST integrates the local feature extraction efficiencies of CNNs with the global context modeling capabilities of transformers, resulting in a model that is powerful yet lightweight, well-suited to a real-time or mobile use case. Additionally, we incorporate Lipschitz continuous attention mechanisms to improve training stability and accelerate convergence. We evaluate FAST on the ADIMA dataset, a multilingual corpus towards real-time profanity and abuse detection, as well as on the more traditional AudioSet. FAST matches or exceeds much larger models on several ADIMA languages and remains competitive on AudioSet, while using two orders of magnitude fewer parameters.**

***Index Terms*—transformers, convolution networks, Lipschitz continuity, abuse detection, audio classification**

I. Introduction

The landscape of audio classification has undergone significant evolution, primarily driven by the adoption of deep learning models that transform raw audio signals into actionable insights [1]–[3]. Traditionally, convolutional neural networks (CNNs) have been at the forefront of this transformation, as their inherent spatial locality and translation equivariance have been effective on audio data represented as spectrogram images [4], [5]. In parallel, the success of architectures in the domain of natural language processing has paved the way for their integration into audio classification tasks, particularly through hybrid models that combine CNNs with self-attention mechanisms to enhance the models' ability to capture long-range dependencies [6]–[10].

However, despite the advancements, the field continues to grapple with the opposing challenges of achieving high performance and optimizing model efficiency—the latter critical for applications requiring real-time processing on resource-constrained devices. Inspired by recent innovations in visual computing such as the Vision Transformer (ViT) [11] and MobileViT [12], which address similar challenges in image processing, we propose a novel architecture: the Fast Audio Spectrogram Transformer (FAST). FAST harnesses the strengths of both CNNs and transformers, integrating them into a unified framework optimized for audio signals.

In creating FAST, we draw inspiration from the MobileViT architecture, which combines the spatial inductive biases of CNNs with the global information processing capabilities of transformers. This synergy allows for a lightweight and efficient model. Moreover, to address the instability issues often encountered during the training of transformer models, we integrate principles from Lipschitz continuity to ensure robustness and stability through the entire training process [13].

This paper introduces FAST, detailing its architecture and the rationale behind combining CNNs with a transformer backbone. We also discuss the integration of Lipschitz continuous components to enhance training stability. The effectiveness of FAST is demonstrated through evaluations on the ADIMA [14] and AudioSet [15] datasets, which showcase its capabilities in multilingual abusive speech detection and sound event detection, respectively. FAST offers a strong accuracy–efficiency trade-off on these tasks while using two orders of magnitude fewer parameters than widely used spectrogram transformers and self-supervised speech encoders.

II. Related Work

Recent advancements in audio classification have increasingly explored transformer-based architectures to enhance both performance and efficiency. One of the pioneering models, the Audio Spectrogram Transformer (AST), was the first to rely solely on self-attention mechanisms without convolutional layers, demonstrating that transformers could achieve state-of-the-art results on benchmarks like AudioSet and Speech Commands V2, challenging the dominance of CNNs [6]. Following this, HTS-AT introduced a hierarchical token-semantic structure, significantly reducing model size and training time while maintaining high accuracy, particularly in audio event detection, thus outperforming earlier CNN-based models in both efficiency and localization [16]. Another key model in this area is Wav2Vec, which leverages unsupervised pretraining on speech data [17]–[19]. Wav2Vec models have been used to extract features from raw audio for various downstream audio tasks, including abuse detection in multilingual corpora like ADIMA [14]. Building on these advances, our proposed FAST architecture integrates the spatial inductive biases of

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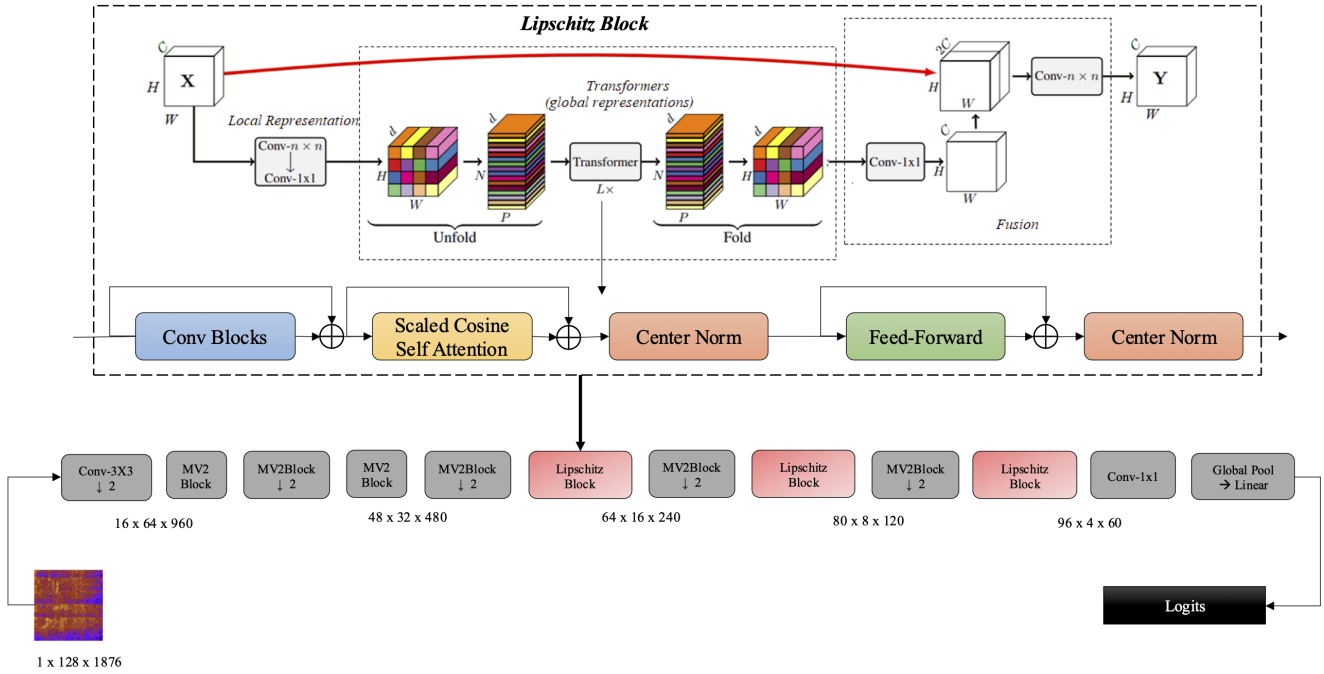


Fig. 1. Architecture of the FAST model, highlighting its combination of CNNs and transformers with Lipschitz continuous attention mechanisms. The upper section illustrates the integration of these components, while the lower section presents the full architecture, including MobileNetV2 blocks and Lipschitz-modified blocks for enhanced efficiency and stability.

CNNs with the global contextual capabilities of transformers while incorporating Lipschitz continuous attention mechanisms for enhanced training stability. FAST aims to strike a balance between performance and computational efficiency, particularly in resource-constrained environments, and in the following sections, we define the architecture and compare its effectiveness against the AST, HTS-AT, and Wav2Vec-based XLSR-53 [19].

III. FAST Architecture

FAST employs an efficient convolutional feature extraction process inspired by lightweight architectures like MobileViT. For a given input audio spectrogram $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, the model applies a 3×3 convolution followed by a point-wise (1×1) convolution, resulting in a transformed tensor $\mathbf{X}_L \in \mathbb{R}^{H\times W\times d}$, where $d > C$. The 3×3 convolution encodes local spatial features, while the point-wise convolution projects the input into a higher-dimensional space by learning linear combinations of the channels:

$$\mathbf{X}_L = \mathbf{X} * \mathbf{W}_{\text{conv3x3}} * \mathbf{W}_{\text{pointwise}}. \tag{1}$$

This setup captures local relationships while preparing the data for global context modeling.

To incorporate global information, $\mathbf{X}_L$ is divided into N non-overlapping patches $\mathbf{X}_U \in \mathbb{R}^{P\times N\times d}$, where $P = wh$ is the number of pixels in each patch. For each patch p, a transformer layer models inter-patch dependencies:

$$\mathbf{X}_G(p) = \text{Transformer}(\mathbf{X}_U(p)), \quad 1 \le p \le P. \tag{2}$$

This operation preserves spatial order within and across patches, allowing each pixel to encode both local and global information. After processing, $\mathbf{X}_G$ is folded back into $\mathbf{X}_F \in \mathbb{R}^{H\times W\times d}$, projected with a 1×1 convolution, concatenated with the convolutional skip, and fused with a 3×3 convolution. This design ensures that the effective receptive field spans the entire spectrogram while maintaining computational efficiency.

A. Lipschitz-Aware Transformer Blocks

A key innovation of FAST lies in its Lipschitz-aware transformer blocks, designed to enhance stability by keeping activations and residual branches bounded [13]. This is achieved through CenterNorm, Scaled Cosine Similarity Attention (SCSA), Weighted Residual Shortcuts (WRS), and spectral-norm initialization of convolution and linear weights.

1) CenterNorm: FAST applies CenterNorm over the spatial axes $\mathcal{S}$ (every dimension except batch and channel). Unlike the mean-only CenterNorm of LipsFormer, the implemented operator also divides by the spatial standard deviation (InstanceNorm with learned affine parameters):

$$\text{CN}(\mathbf{x}) = \boldsymbol{\gamma} \odot \frac{\mathbf{x} - \boldsymbol{\mu}}{\sqrt{\boldsymbol{\sigma}^2 + \epsilon}} + \boldsymbol{\beta}, \tag{3}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}^2$ are the mean and population variance of $\mathbf{x}$ over $\mathcal{S}$, $\epsilon = 10^{-6}$, and $\boldsymbol{\gamma}$, $\boldsymbol{\beta}$ are learnable.

2) Scaled Cosine Similarity Attention (SCSA): Queries and keys are ℓ_2-normalized along the head dimension; values are not. After a CenterNorm, the projections are

$$\mathbf{q}_i = \frac{\mathbf{x}_i^\top \mathbf{W}^Q}{\|\mathbf{x}_i^\top \mathbf{W}^Q\|_2}, \quad \mathbf{k}_i = \frac{\mathbf{x}_i^\top \mathbf{W}^K}{\|\mathbf{x}_i^\top \mathbf{W}^K\|_2}, \tag{4}$$

$$\mathbf{v}_i = \mathbf{x}_i^\top \mathbf{W}^V. \tag{5}$$

With head width d_h and a fixed temperature $\tau = 10$,

$$\mathbf{P} = \mathrm{softmax}\big(\,\mathrm{clip}(\tau d_h^{-1/2} \mathbf{Q}\mathbf{K}^\top, -50, 50)\big), \tag{6}$$

$$\mathrm{SCSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{P}\mathbf{V}. \tag{7}$$

Clipping the cosine logits is a numerical guard. A linear projection maps the concatenated heads back to the embedding width.

3) Weighted Residual Shortcuts (WRS): Each residual branch is scaled by a per-channel vector $\boldsymbol{\alpha}$ (initialized to 0.1) and stochastic depth, then CenterNorm is applied to the sum:

$$\mathrm{WRS}(\mathbf{x}, f) = \mathrm{CN}\left(\mathbf{x} + \mathrm{DropPath}(\boldsymbol{\alpha} \odot f(\mathbf{x}))\right). \tag{8}$$

The same pattern is used in identity MobileNetV2 blocks (stride 1, matched channels).

4) Lipschitz Transformer Layer: Attention and the MLP each use WRS. The MLP is CenterNorm, a linear layer, SiLU, dropout, and a second linear layer. Convolution and linear weights are divided at initialization by a power-iteration estimate of their spectral norm, which bounds the initial Lipschitz constant of each map. Figure 1 illustrates the Lipschitz block.

B. Overall Architecture

The trunk of the FAST model alternates between inverted residual convolutions (MobileNetV2-style expand–depthwise–project with SiLU and squeeze-and-excitation) and Lipschitz-aware transformer blocks. Convolutional layers reduce spatial size while capturing local features; transformer blocks integrate global context over unfolded spectrogram patches. After a 1×1 projection of the transformer path, features are concatenated with the convolutional skip and fused by a 3×3 convolution.

At the final stage, a 1×1 convolution with SiLU is followed by global average pooling and a linear classifier.

IV. Experiments

A. ADIMA Dataset

1) Dataset Description: The ADIMA dataset contains 11,775 audio samples (65 hours) in 10 Indic languages (Hindi, Bengali, Punjabi, Haryanvi, Kannada, Odia, Bhojpuri, Gujarati, Tamil, and Malayalam) spoken by 6,446 unique chatroom users. The accompanying task is one of classifying audio samples as "abusive" or "not abusive". The dataset is well balanced between the two classes - 43.48% of samples are "abusive". The samples of each language are split randomly 70-30 to form the training and test sets. We use the official ADIMA train/test splits from [14].

TABLE I
PERFORMANCE COMPARISON OF XLSR-53, AST, HTS-AT, AND FAST ACROSS DIFFERENT LANGUAGES (ACCURACY/F1 SCORE), UNDER THE CORRECTED EVALUATION PROTOCOL OF SECTION IV-A2. BOLD INDICATES WHERE FAST SURPASSES OTHER MODELS.

Language	XLSR-53 (Acc/F1)	AST (Acc/F1)	HTS-AT (Acc/F1)	FAST (Acc/F1)
Hindi	0.77, 0.77	0.76, 0.75	0.77, 0.76	**0.80, 0.81**
Bengali	0.81, 0.79	0.77, 0.80	0.76, 0.80	0.81, **0.84**
Punjabi	0.80, 0.82	0.85, 0.85	0.83, 0.83	**0.87, 0.87**
Haryanvi	0.82, 0.79	0.81, 0.80	0.82, 0.81	**0.83, 0.82**
Kannada	0.83, 0.79	0.81, 0.79	0.79, 0.78	0.79, 0.72
Odia	0.83, 0.82	0.80, 0.79	0.80, 0.79	0.82, 0.79
Bhojpuri	0.76, 0.71	0.79, 0.80	0.80, 0.81	0.78, **0.83**
Gujarati	0.79, 0.69	0.82, 0.72	0.83, 0.73	0.78, 0.68
Tamil	0.80, 0.75	0.82, 0.79	0.81, 0.79	0.75, 0.63
Malayalam	0.81, 0.75	0.79, 0.76	0.79, 0.78	**0.84**, 0.76

2) Evaluation Protocol: Input spectrograms are 128-bin log-mels at 22.05 kHz with hop length 512, cropped or padded to 1024 frames (≈23.8 s). After scaling each clip to $[-1, 1]$, silence is padded with -1, not 0. Zero-padding after this normalization looks like mid-energy to global average pooling and inflates scores on languages with shorter or more variable clips (notably Kannada, Gujarati, and Tamil in an earlier version of this paper). At test time we average logits over overlapping windows with stride $0.5\times$ the window width. Table I uses this protocol.

3) Training and Results: We evaluate the audio classification performance of FAST against the AST, HTS-AT and XLSR-53 models. Table I presents comparative results of FAST against other models across various languages in terms of accuracy and F1-scores.

TABLE II
COMPARISON OF MODEL EFFICIENCY. FEWER PARAMETERS AND QUICKER INFERENCE TIME TOGETHER INDICATE BETTER EFFICIENCY.

Model	Parameters	Inference Time (secs)
XLSR-53	300M	0.0579
AST	87M	0.0967
HTS-AT	31M	0.0231
FAST (Ours)	**2.78M**	**0.0194**

Table II provides a comparative overview of the parameter counts and inference latencies for leading audio classification models. Number of parameters is a crucial metric reflecting the complexity and computational requirements of the model architecture. Inference time is key to real-time usability and mobile device energy consumption, and smaller models do not necessarily result in faster inference. For inference time testing, we follow the single stream scenario discussed in MLPerf Inference Benchmark [20].

We find that FAST competes with much larger models such as XLSR-53, AST, and HTS-AT, despite a more streamlined design with faster inference and two orders of magnitude fewer parameters. Under the corrected evaluation protocol (Section IV-A2), FAST is best on both accuracy and F1 for Hindi, Punjabi, and Haryanvi, best on F1 for Bengali and

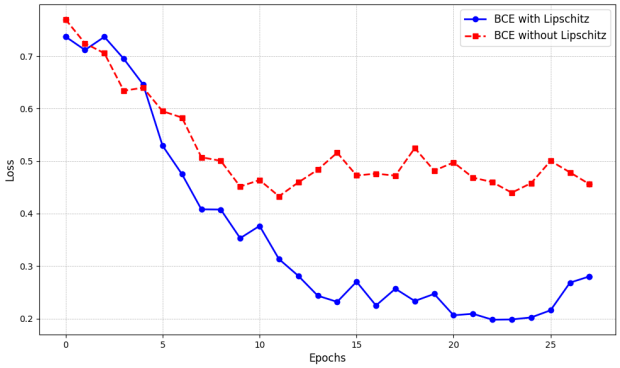


Fig. 2. Comparison of training stability and efficiency in FAST with and without Lipschitz continuity components on the ADIMA Hindi language set. Loss is measured with binary cross-entropy (BCE).

Bhojpuri, and best on accuracy for Malayalam. Kannada, Gujarati, and Tamil are weaker on F1, reflecting a precision–recall trade-off on Yes-minority splits rather than a change in model family.

These results demonstrate FAST's potential in deploying effective audio classification solutions at a fraction of the standard complexity, which may be game-changing for resource-constrained environments such as ADIMA's intended mobile or edge devices. The model's performance across various languages establishes it as a robust option for real-world applications.

4) Effect of Lipschitz Components: We also observe that the integration of Lipschitz continuity components into the FAST architecture benefits training stability and efficiency. During experiments, we found that inclusion of Lipschitz constraints mitigated instability issues in our deep model. Moreover, training pace was accelerated compared to the version of our model without this component — Lipschitz regularization appeared to bring quicker convergence, reducing computational overhead and resource requirements. Incorporation of Lipschitz continuity components within FAST not only bolstered its robustness but also streamlined its training dynamics. Figure 2 illustrates the impact of integrating Lipschitz components on training in the FAST architecture for Hindi language. The plot is similar across all languages.

B. AudioSet

AudioSet [15] is a large-scale dataset comprising over 2 million 10-second audio clips sourced from YouTube, each annotated with one or more of 527 possible sound labels. The dataset is organized into three sets: a balanced training set of 22k samples, a full training set containing 2 million samples, and an evaluation set with 20k samples. For our experiments on AudioSet, we adhere to the same training pipeline as described in [6], ensuring consistency with prior work.

To evaluate the performance of our model, we compare FAST with the state-of-the-art transformer-based models AST and HTS-AT. Table III presents the mean average precision (mAP) scores for each model. While AST and HTS-AT achieve mAP scores of 0.459 and 0.488, respectively, FAST achieves a slightly lower mAP of 0.448. Despite this difference, FAST maintains competitive performance, particularly considering its lightweight architecture. This underscores the potential of FAST for real-time applications where efficiency and reduced model complexity are critical.

TABLE III
COMPARISON OF MAP SCORES FOR AST, HTS-AT, AND FAST ON AUDIOSET.

Model Architecture	mAP
AST	0.459
HTS-AT	0.488
FAST (Ours)	0.448

TABLE IV
MODEL CONFIGURATION FOR FAST.

Parameter	Value
Image Size	(128, 1024)
Hidden Dimensions	(96, 128, 144)
Channels	(16, 32, 48, 48, 64, 64, 80, 80, 96, 96, 384)
Number of Classes	N
Expansion Ratio	4
Kernel Size	3
Patch Size	(2, 2)
Depths	(2, 4, 4)

C. Input and Hyperparameter Details

For ADIMA, we pretrain FAST for 15 epochs on the union of all 10 language training sets, then fine-tune for 20 epochs per language from that checkpoint. We use AdamW with learning rate 4×10^{-4} (fine-tune 1.6×10^{-4}), weight decay 1×10^{-4}, batch size 16, class-weighted cross-entropy with label smoothing 0.05, cosine decay with a 2-epoch warmup, and an exponential moving average of weights with decay 0.995. Checkpoints are selected by Acc+F1 on the official test split. The primary metrics are accuracy and F1 on the abusive class. In experiments on AudioSet, the model is trained for 5 epochs with an initial learning rate of 1e-5, halved after the second epoch, batch size 16, and using mean average precision (mAP) as the primary metric. The model configuration is shared in Table IV.

V. CONCLUSION

In this paper, we presented FAST, a novel and lightweight architecture that combines convolutional neural networks with a transformer backbone, augmented by Lipschitz-based enhancements, for audio classification tasks. Our model demonstrates competitive performance on both the ADIMA and AudioSet tasks while using two orders of magnitude fewer parameters than common spectrogram transformers. Motivated by the "live chatroom safety" origins of ADIMA, we envision FAST to be suitable for a real-time, on-device setting due to its small size and low latency.

For future work, we plan to investigate the potential of transfer learning from vision models to further enhance generalizability to the broader realm of audio classification tasks. Additionally, preliminary experiments suggest that FAST holds promise for few-shot learning across languages using meta-learning techniques like MAML, offering new directions for improving cross-lingual audio tasks.

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